



Semester 1 Examinations 2016

Exam Code(s) 1SD1, 1MDM1, 1IF1, 1MF1, 1MCR1, 1SD3
Exam(s) Higher Diploma in Software Design and Development
Higher Diploma in Software Design and Development,
Industry Stream
Masters in Information Technology
Masters in Software Design and Development
Masters in Digital Media
Masters Clinical Research

Module Code(s) CT511
Module(s) Database Development

Paper No.
Repeat Paper

External Examiner(s) Prof. Liam Maguire
Internal Examiner(s) Prof. Peter Mc Hugh
Dr. Jim Duggan
*Dr. Attracta Brennan

Instructions: Answer **questions 1 and 2** and 2 other questions
All questions carry equal marks

Duration 2 Hours
No. of Pages 6
Discipline(s) Information Technology
Course Co-ordinator(s)

Requirements:

MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper

Other Materials

Release to Library: Yes ☒ No ☐

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Question 1

- a) Create an **EERD** for the following:

A Department can be either Arts, Financial or HR. As of yet, the college has no plans to add on new departments but may in the future. Departments are identified by the Department ID, Head and Phone number. Employees are assigned to only one department. The Arts department has an offsite office with a dedicated landline. The Financial department has an external auditor – his name and contact details are captured. Meanwhile the HR department has its own specific opening hours. A department has at least one employee (identified by ID, name, address and date of birth) working in it. There are different types of employees. They must be either hourly, salaried or consultant. Some employees are assigned to a couple of projects. Other employees have no projects allocated to them. Regardless of whether or not an employee has a project allocated to them, every project has at least one employee working on it.

Various pieces of data are kept on each project. They include; description, date started, estimated cost, project code and total cost to date. Hourly employees are differentiated by monthly wages and their hourly rate. Salaried employees differ based on their annual salary and the stock options afforded to them.

Meanwhile a consultant has a daily rate and an assigned contract number. The consultants generally hold many clinics on a weekly basis. Rules stipulate that a clinic has to have at least one consultant present in order to operate. A project must be either NGO, state or private.

(16 marks)

- b) Discuss the history of databases, outlining strengths and limitations of the various approaches where relevant.

(7 marks)

- c) What is meant by the term meta-data (use an example to show your understanding)?

(2 marks)

Question 2

- (a) Write the SQL for a MS SQL Server DB to :

1. Create a table called employee with the following data; employee id (integer), first name, last name, address and title can be up to 50 characters in length, age, salary can be any 5 figure digit. Set the employee id as the primary key.
2. Insert the following data into your table; 1, Luke Duke whose age is 45 and who lives at; 2130 Boars City is a programmer with a salary of 11000.
3. Insert the following data; 2, Jonie Weber, Secretary, 28, 19500; 3, Patsy Weber, Programmer, 32, 45300; 4, Dirk Smith, Programmer II, 45, 75020; 5, Tom Jones, Secretary, 32, 81000.
4. Select all columns for everyone with a salary over 30000.
5. Select all first and last names for everyone that's under 30 years old.
6. Select first name, last name, and salary for anyone who is just a "Programmer".
7. Select all columns for everyone whose last name contains "ebe".
8. Delete all entries for anyone whose first name is Jonie or Tom.
9. Delete the entire table completely inclusive of all entries.
10. Show only distinct ages.
11. Show the average salary of all programmers.
12. Count the number of programmers.

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13. Select the maximum salary grouped by title.
14. Select the average salary grouped by title where the average salary is greater than 25000.
15. Select all data for those whose salary is greater than 65000 and who are secretaries.
16. Select first name, last name and salary for those whose last name is in the following list; Hernandez, Weber, Timmons.

Given this new table called **ORDER**,

ORDER

OrderId	Emp-Id	Item	Price
O1	1	snowboard	45.00
O2	1	snow shovel	35.00
O3	2	gloves	15.00
O4	3	lantern	35.00
O5		tent	85.00
O6		Snow towel	90.00

17. Write the SQL (join) which displays the employees' first and last name and the items they've purchased, inclusive of price.
18. Write the SQL (join) to show ALL orders. Note : some, orders will not be associated with any employees. Display order id, employee first and last names, item and price.

(20 marks)

(c) Explain the differences between a DBMS and a database. Use examples to show your understanding.

(5 marks)

Question 3

- (a)** Bring the **BOOK** table (overleaf) through 2NF and 3NF. Ensure to go through each of the 5 steps for 2NF. Go through each of the resultant tables arising from the 2NF process and say whether they are/are not in 3NF. Explain your reasoning. If they are not in 3NF, draw the Functional Dependencies Diagram and map to relations.

(12 marks)

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BOOK

Author	Nationality	Book title	Genre	Number of pages	Section
William Shakespeare	English	The Comedy of Errors	Comedy	100	A
Markus Winand	Austrian	SQL Performance Explained	Textbook	200	B
Jeffrey Ullman	American	A First Course in Database Systems	Textbook	500	B
Jennifer Widom	American	A First Course in Database Systems	Textbook	500	B

- (b) After proceeding through 3NF, it was later found (when reviewing the data from the **STUDENT** table below) that all Offering# in combination with the student number (S_Num) were reflective of the Review Date for the thesis submission. Bring the STUDENT table through Boyce Codd Normal Form (BCNF) by; (a) drawing the Functional Dependency Diagram, (b) mapping to relations and (c) redrawing the resultant tables.

STUDENT

S_Num	Lecturer Code	Offering#	Review Date
123599	FIT104	01764	2nd March
123599	PIT305	01765	12th April
123599	PIT107	01789	2nd May
346700	FIT104	01764	3rd March
346700	PIT305	01765	7th May

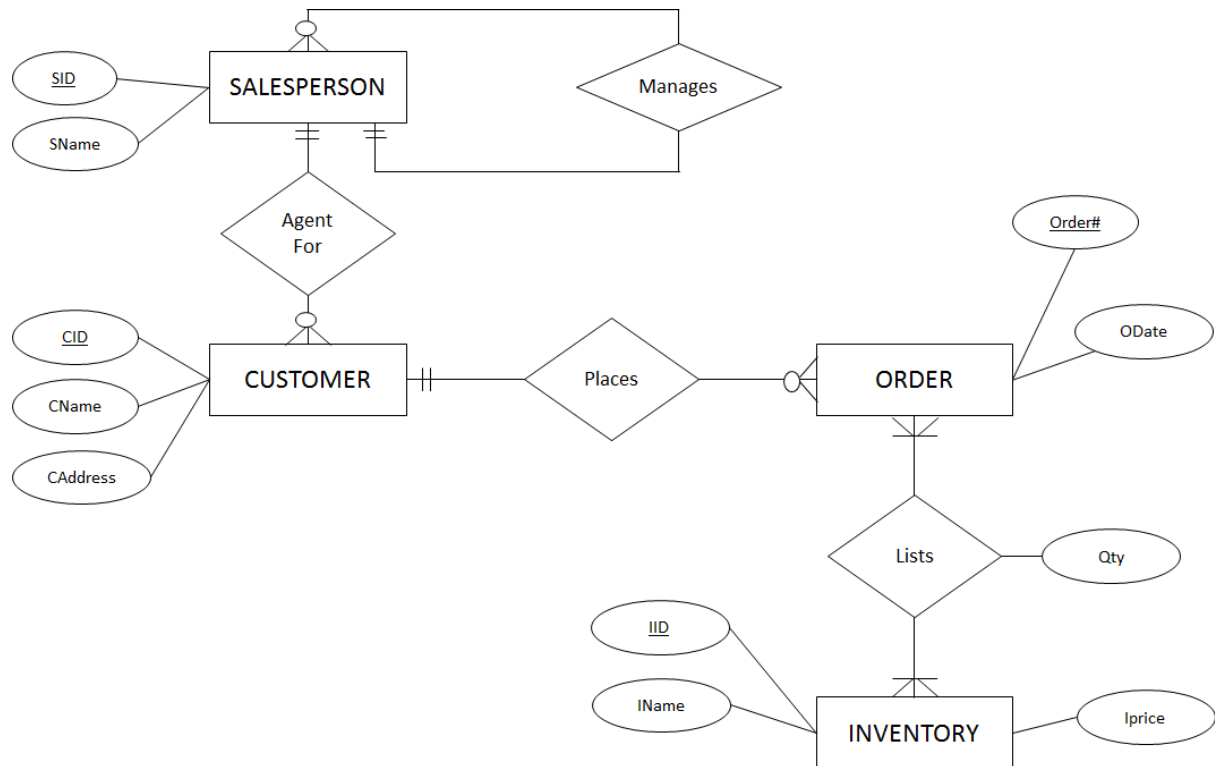
(9 marks)

- (c) Discuss the 4 integrity constraints. Use examples to show your understanding.

(4 marks)**PTO**

Question 4

a) Map the following to relations (use the short hand approach):



(10 marks)

b) Compare and contrast Normalization and Denormalisation (outlining strengths and weaknesses of each). Use examples to show your understanding.

(10 marks)

c) Compare and contrast Horizontal and Vertical partitioning. Use examples to show your understanding.

(5 marks)

Question 5

a) Describe the 3 options for Denormalisation. Use examples to show your understanding.

(12 Marks)

b) Discuss what is meant by Distributed Database Management Systems, outlining their advantages and disadvantages.

(9 Marks)

c) Explain what is meant by the terms 'Data Dependence' and 'Data Abstraction'. Use examples to show your understanding.

(4 Marks)

Question 6

a) Draw and then discuss the 'Three Schema Architecture'

(10 Marks)

b) Given the following table called Employee

Employee

EmployeeID	FirstName	LastName	Age	Title	Salary
1	Luke	Duke	45	Doctor	90000
2	Anna	Lyons	34	Doctor	85000
3	Mike	Mullins	42	Nurse	48000
4	Sarah	Jones	28	Nurse	32000
5	Marian	Murphy	29	Nurse	34000

Write the SQL for MS SQL Server DB to :

- Find the max salary of the job titles using group by. Display title as *Job Title* and salary as *Maximum Salary*.
- Find the total amount of salary paid to each of the job titles using group by. Display title as *Job Title* and salary as *Total Salary Paid*.
- Count the number of people in each job category (title) where there is more than two people in the job category (i.e. title) – use count, group by and having. Display title as *Job Category* and the count of the numbers who have the same title as *Count of People in this Category*.
- Count the number of people in each job category (title) where the salary is more than 30000 and where there the count of the number of people in this job category is 2 or more – use count, group by and having. Display the count as *Count of People in this Category with salaries greater than 30000 and where there two or more people in this department*

(10 Marks)

c) Compare and contrast a Data Warehouse with a Transactional Database, outlining how and why they would be used.

(5 Marks)